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Man-made Measures: Using AI to diagnose and treat Parkinson's Disease

I. Introduction

90,000 people are diagnosed with Parkinson's Disease in the United States each year. It is the fastest growing neurological disease in the world, and the number of cases globally is expected to reach 12 million by 2040. **Artificial Intelligence may be used to advance and hasten research to diagnose and cure Parkinson's.** Although there are many social, ethical, and legal obstacles that need to be overcome, and the fact that AI is not fully sophisticated, due to its now capable data analysis abilities, it has the potential to be more effective paired with human experts in chronic disease research, particularly neurodegenerative diseases such as Parkinson's.

II. Background

Parkinson's is a neurodegenerative disease caused by the deterioration of dopaminergic neurons in a patient's brain, manifesting symptoms such as bradykinesia, tremors, rigidity etc. AI, or Artificial Intelligence, are systems that mimic human intelligence, ability to learn, problem-solving, pattern recognition, and decision-making. Since its discovery in 1850, Parkinson's Disease, or PD, has not had a cure. Experimental treatments exist, however none are good enough to be effective consistently. AI can access the entire internet to solve problems, making it able to elevate and augment

clinicians care (Jackson). It can also analyze huge datasets very quickly and help clinicians make more timely decisions on how to carry out diagnosis or treatment, saving hours and maybe days of research (Ellis). AI has demonstrated accuracy in diagnosing and detecting PD, able to detect abnormalities in gait, handwriting or facial expression (Wu; Yang). AI has also shown proficiency in data analysis and predictions in the fields of possible treatments for PD, such as stem cell therapy and gene therapy.

III. Possible treatments

A. Stem cell therapy

1. Cells that can self-repair and differentiate into different types of cells and take on their function (Choudhery).
2. Shows promise
3. AI needs large and comprehensive datasets in order to be truly effective
 - a) There is bias in datasets, and therefore AI needs larger and more comprehensive ones to be most effective. In other words it needs to be trained on data from an entire population, a large range of people to give AI unbiased training (Matheny). Being too general can have consequences, and learning will fail across demographics, creating inequalities due to biased training (Minesh).
 - b) Data collection is crucial. AI needs to be trained using supervised learning, and then optimized using a human must be done to streamline the model, making it capable of these advanced processes (Dixit; Saxena).

- c) Small datasets, which lead to Inept models that aren't trained properly, can then lead to inaccurate diagnosis (Yang).

4. The black-box problem

- a) Refers to the inability of humans to understand AI's decision-making process and extract reasons for why and AI did what it did.
- b) Creates a lack of trust in professionals.
- c) The main way to fix this problem is to have a human in the loop, monitoring the AI and making sure it doesn't hallucinate, at least for the foreseeable future, because AI isn't capable of advanced processes yet. Use red teaming, a process in which, over a period of a few months, the AI is given all kinds of "Doomsday" scenarios to see how it responds, and whether that response is what the team wants and expects. Using guardrails and interpretable models also helps (Saxena). Making trustable AI models is also possible.

5. Further research of stem cells. There are differences between potency and function, depending on sourcing, harvesting method, and delivery. There are also obstacles in terms of scaling up and standardization (Mukherjee).

B. Gene therapy

- 1. AI has shown remarkable progress and capability in rewriting the gene (Dixit).
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IV. There are ethical concerns regarding using AI in medicine

A. Data Privacy

1. Data security and privacy protection. You need to prevent AI from releasing this information to unauthorized individuals and also prevent it from accessing the internet (Saxena).
2. It can be done by using guardrails to curb AI's creativity and power, redacting information that the AI does not need to know; the AI needs to be treated like any other human employee, using regular audits and checks (Saxena).
3. Data "stewardship," using secure storage and encrypted transmissions, restricting access to information. Participants and patients should also be informed of what AI entails and give educated consent (Martinez-Martin).
4. Data diversity also matters here, as bias in the AI model is an ethical concern as well as a quality concern. AI needs to be trained through supervised and third-party learning to have minimal bias (Martinez-Martin).

B. AI is known to perform unsolicited and uninformed diagnosis (Youngmann).

1. Responsibilities extend to technology companies from clinicians when implementing AI in the medical workplace (
2. Youngmann).

V. Conclusion

A. Solutions to the problems that hinder AI are already making their way into the mainstream, and very soon the implementation of AI in medicine will take place. AI's exceptional pattern-recognition, decision-making, and learning skills make it a primary augmentation for clinicians, leading to precise diagnosis and treatment of conditions and diseases previously impossible to address. Despite the many obstacles that need to be overcome, including that AI is not fully sophisticated. But due to its now capable data analysis abilities, it has the potential to be more effective paired with human experts in chronic disease research. In the next decade, a treatment for Parkinson's can become effective and mainstream, paving the way for other conditions and diseases previously thought to be incurable, to also be treated.

VI. Works Cited

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